

Julian Henry

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Work Experience

Senior Full-Stack Engineer

October 2024 – Present

Macquarie Financial Ltd. — Houston, TX · Full-time

- Deploy Java servlet services for high-priority trading paths across London, São Paulo, and Sydney.
- Port as-dealt deal logic from C++ to Python for ASIC regulatory reporting and trade exposition.
- Co-author a library adding risk and workflow controls to trades across 100+ financial products.
- Ship C++17 enhancements in a 1M+ line trading platform for low-latency, high-availability markets workflows.

Software Engineer

May 2023 – October 2024

Allis-Gleaner Corporation (AGCO) — Atlanta, GA · Full-time

- Hardened AWS Kafka IoT telemetry for a connected agriculture fleet across 140 countries.
- Built RESTful APIs for hardware device orchestration that cut integration test cycles by 500%+.
- Shipped Angular features on Fendt.One precision agriculture for 10,000+ remote farm operations.

Data Engineer

August 2020 – August 2022

SimSpace Corporation — Boston, MA · Full-time

- Stress-tested ML threat-detection models via adversarial red-team simulations before production.
- Built cybersecurity analytics pipelines with NLP that improved threat detection rates by 72%.
- Led performance work on a 30,000+ line Python/Pandas ETL pipeline, delivering 10× throughput gains.

Assets & Liabilities Intern

June 2019 – May 2020

Transamerica Corporation — Baltimore, MD · Internship → part-time

- Validated actuarial risk models for variable annuity portfolios with NumPy, scikit-learn, and pandas.
- Reached 99.7% simulation accuracy on reconciliation and stress testing for a \$6B+ AUM book.

Technical Skills

Languages: C++17, Python, Java, TypeScript, JavaScript, SQL, Julia, Rust, WebAssembly

Frontend: React, Next.js, Angular, HTML5, CSS3

Backend: Spring Boot, Node.js, Express, REST APIs, Java Servlets

Cloud / Infra: AWS, Kafka, Docker, Linux, GitHub Actions, CI/CD, Cloudflare

Databases: PostgreSQL, MongoDB, DynamoDB, Neo4j, SurrealDB, SQLite, FAISS, Vector DBs

Data / ML: Pandas, NumPy, scikit-learn, TensorFlow, PySpark, RAG, Tableau

Education

Bachelor's in Applied Science, University of Pennsylvania

- Vice President, Artificial Intelligence Society
- Undergraduate thesis on differential privacy under Rajeev Alur, published with distinction

Certificates

DELE B2 (Instituto Cervantes) — official upper-intermediate Spanish credential

Open Source & Portfolio

Founder & Independent Developer

- 1wg.ai — AI global news gazette with RAG over a vector database
- glottosphere.com — 200+ language dictionary with FAISS similarity search
- [GitHub](#) / PyPI — [ghee](#), [tatuagem](#), [readme-rosetta](#)
- **Selected builds:** [intersekt](#), [quantum-plankton-ml](#), [laciyo](#)